Some applications of Artificial Intelligence

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Overview

- I will give an overview of AI with a focus on deep learning via neural networks.
- The examples will be from MSc projects: classifiers for churn and medical images.
- At the end, I will briefly discuss the issues with Artificial General Intelligence (AGI), because this is a current focus of the US big technology companies.
- I will review the crisis in pure mathematics from AI.



- The term **Artificial Intelligence** was first used in a Dartmouth workshop in 1956.

  An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves.



Symbolic AI

For most of the history of AI (mid 1950s to mid 1990s), symbolic AI was the main area of AI.

- There are some basic facts.
- Human experts have to write the basic facts.
- There are rules to generate new facts, using for example types of logic.
- Computer languages such as prolog and lisp are used for symbolic AI.
- Basis of expert systems.

It has been found that it is difficult to write enough facts to build a large AI system.

Example of a prolog program

```
% --- FACTS ---
% parent(Parent, Child)
parent(john, mary).
parent(john, tom).
parent(mary, alice).
parent(mary, bob).
% --- RULES ---
% X is a grandparent of Z if X is a parent of Y, and Y is a
grandparent(X, Z) :-
    parent(X, Y),
    parent(Y, Z).

?- grandparent(john,alice).
true .
```

Machine learning

The algorithms learn from the data.

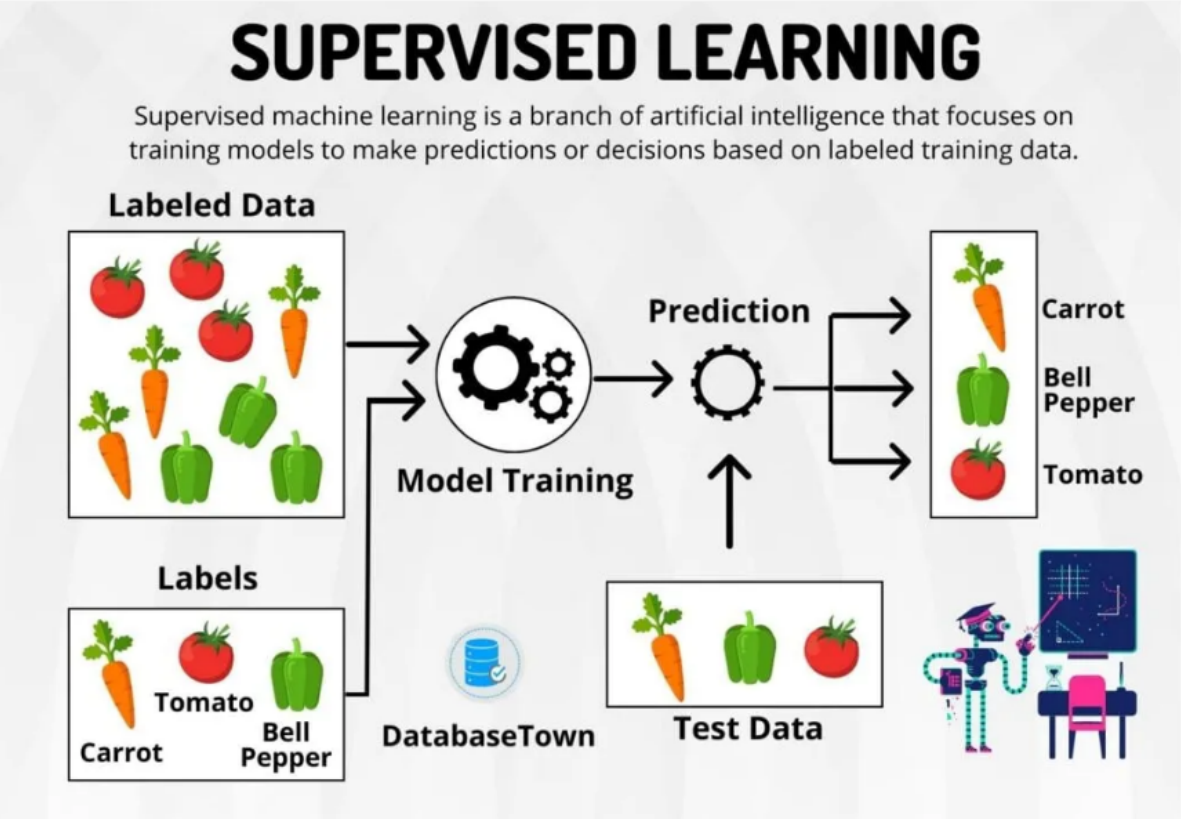
- The algorithms have many free parameters.
- The free parameters are fixed by minimizing the difference between the output of the algorithm and data.

Examples

- Support Vector Machine.
- logistic regression.
- Neural Networks.

Machine learning is a subset of AI.

Supervised learning



Introduction to Neural network

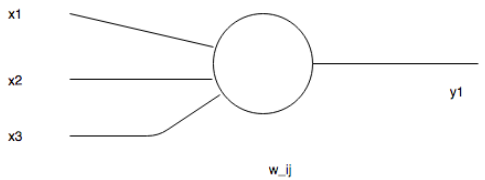
- A neural network is a machine learning algorithm
- A neural network is a universal function approximator.
- They are used in many applications:
  - Classification of text, images, speech (translation)
  - Regression.
- The original motivation for the development Neural Network came from models of the brain. To my mind this makes the construction a little strange. (I will discuss this later.)

Basic Neural network

The neural network has some input nodes (an image) which connect to output nodes (is it a neural network).
The representation of a single layered neural network is

$$y_k = \phi(\sum_{i}^{N} w_{ki} x_i)$$

- x_i are the input values
- y_k are the output values
- w_{ki} are the weights. These are determined by the algorithm.
- ϕ is the activation function. (For example a step function.)



Where did neural networks come from?

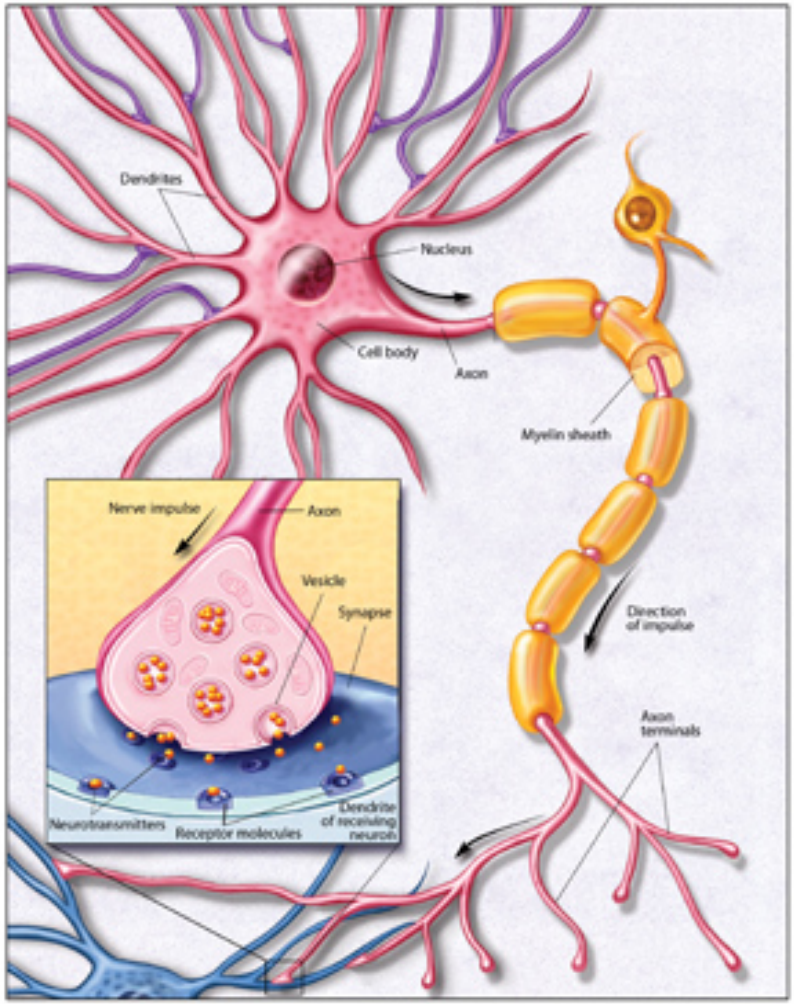
- The original motivation for neural networks came from the modeling the brain.
- This is no longer the main motivation.

From `http://www.brainfacts.org/Brain-Anatomy-and-Function/Anatomy/2012/The-Neuron`

- A mammalian brain contains between 100 million and 100 billion neurons, depending on the species.
- Each mammalian neuron consists of a cell body, dendrites, and an axon.
- The connections between neurons are not static, though, they change over time. The more signals sent between two neurons, the stronger the connection grows.

Neurons in the brain

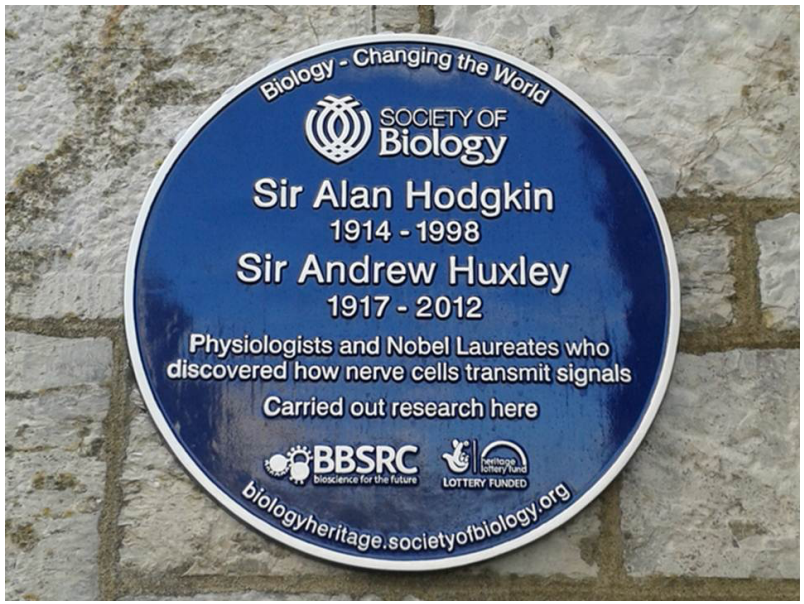
From `http://www.brainfacts.org/Brain-Anatomy-and-Function/Anatomy/2012/The-Neuron`



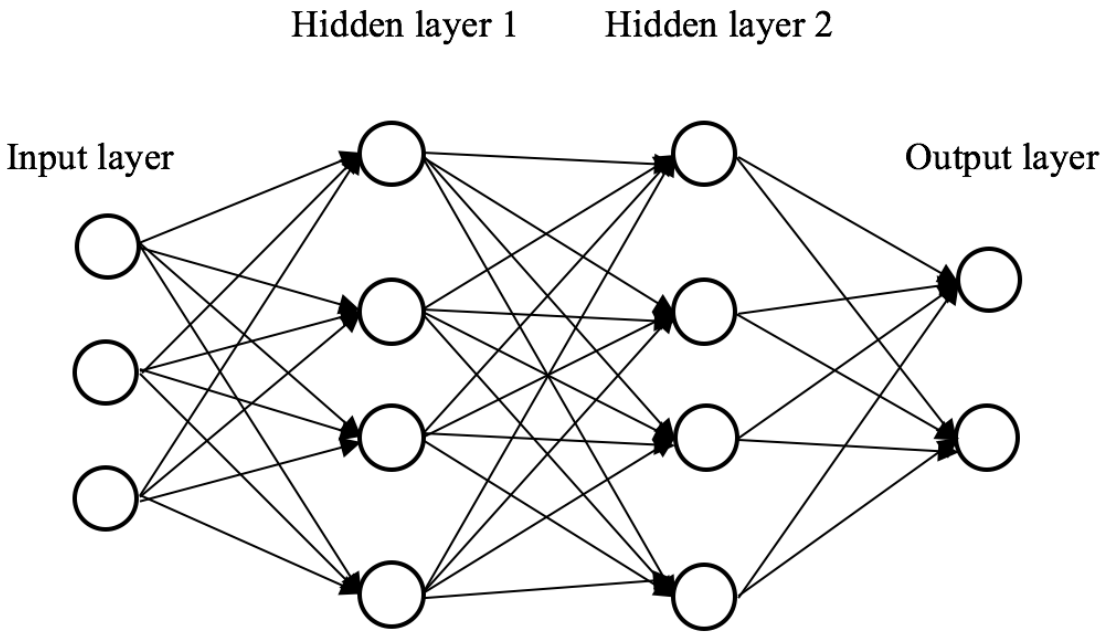
Some Plymouth History

- In 1939, Alan Hodgkin and Andrew Huxley traveled from Cambridge University worked at the Laboratory of the Marine Biological Association in Plymouth to work on the giant nerve fibre of the squid Loligo.
- They were able to record the potential difference (50 mV) between the interior and exterior of the fibre of the squid.
- A share of the Nobel Prize for Physiology or Medicine in 1963 followed.

`https://pmc.ncbi.nlm.nih.gov/articles/PMC3424716/`



Hidden layers



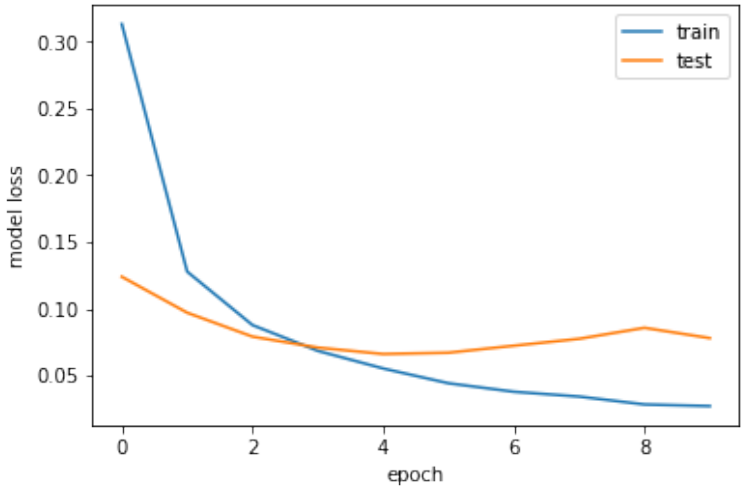
- There are many different types of network. For example convolutional networks.

Training the neural network

The parameters of the neural network are found by optimizing the difference between the algorithm predictions and the data.

Training data Typically use 80 % of the full data set to train the neural net.

Test data Test the neural network on the remaining 20% of the data set. This checks that the algorithm doesn't fit noise.



History of Neural networks

1943 McCulloch and Pitts' (1943) dynamical rule for neurons.

1957 Perceptron (single layer neural network) invented by Frank Rosenblatt.

1969 Book by Marvin Minsky and Seymour Papert (MIT) showed that the Perceptron could not learn XOR logic function. This killed off work on neural networks.

1986 paper by Rumelhart, Hinton, and Williams, titled "Learning Representations by Back-Propagating Errors." This allowed bigger networks to be trained.

1989-1992 During his PhD at Edinburgh University, Craig McNeile went to many seminars on neural networks, but he didn't pay attention.

2006 Rebranded as deep learning

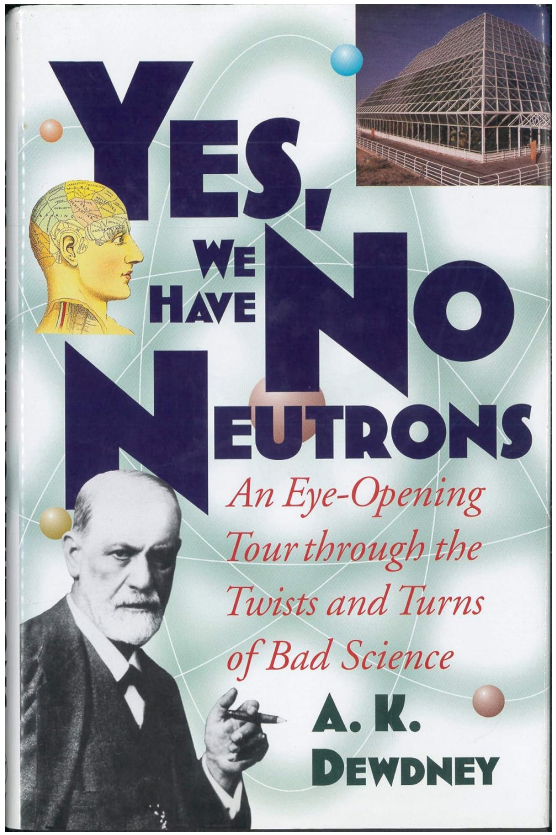
Champion of neural networks

Geoffrey Hinton CC FRS FRSC (born 6 December 1947) is an English Canadian cognitive psychologist and computer scientist, most noted for his work on artificial neural networks. In 2024, he was awarded the Nobel Prize in Physics.



2nd AI winter

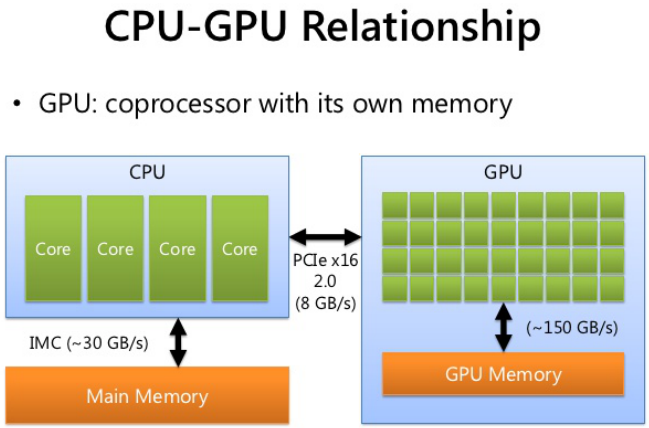
Book that mocks neural network research in the 1990s.



Why did neural networks dominate AI after 2012

- Much bigger data sets were available.
- Graphical Processing Units (GPUs) allowed much bigger networks (with more parameters to be trained.)
- There were competitions for image classifiers that neural networks regularly won after 2012.

Large Scale Visual Recognition Challenge (ILSVRC). It classifies images into 1,000 distinct object categories. 1.2 million images.



Data Science and Business Analytics



- One year Data Science and Business Analytics MSc, at the University of Plymouth, which ends with a 4 month research project.
- This is a conversion MSc with students with no maths or programming background.

Churn

- Customer churn occurs when a customer stop using services provided by a company.
- Some examples:
  - Person stops using a credit card.
  - A person stops using a streaming service, such a Netflix.
  - Meal-kit companies, such as, helloFresh has almost 100% customer churn in a year.
- It is more efficient to keep existing customers than to get new customers.

We want to build a machine learning model to predict customer churn. A company can then give offers to try to persuade customers to stay with them.

Dataset

Use a public tabular dataset stored in a csv (Excel file) that has data on customers that stop using a bank.

Geography	Gender	Age	Tenure	Balance	NumOfProducts	HasCrCard	IsActiveMember	EstimatedSalary	Exited
France	Female	42	2	0	1	1	1	101348.88	1
Spain	Female	41	1	83807.86	1	0	1	112542.58	0
France	Female	42	8	159660.8	3	1	0	113931.57	1
France	Female	39	1	0	2	0	0	93826.63	0
Spain	Female	43	2	125510.82	1	1	1	79084.1	0
Spain	Male	44	8	113755.78	2	1	0	149756.71	1
France	Male	50	7	0	2	1	1	10062.8	0
Germany	Female	29	4	115046.74	4	1	0	119346.88	1
France	Male	44	4	142051.07	2	0	1	74940.5	0
France	Male	27	2	134603.88	1	1	1	71725.73	0
France	Male	31	6	102016.72	2	0	0	80181.12	0
Spain	Male	24	3	0	2	1	0	76390.01	0
France	Female	34	10	0	2	1	0	26260.98	0
France	Female	25	5	0	2	0	0	190857.79	0
Spain	Female	35	7	0	2	1	1	65951.65	0
Germany	Male	45	3	143129.41	2	0	1	64327.26	0
Germany	Male	58	1	132602.88	1	1	0	5097.67	1
Spain	Female	24	9	0	2	1	1	14406.41	0
Spain	Male	45	6	0	1	0	0	158684.81	0

We want to build a machine learning model that uses the variables: CreditScore,Geography,Gender etc., to predict the variable **Exited** (that the customer stops using the credit card).

Heuristic rules for bank customer churn

Surname The surname of a customer has no impact on their decision to leave the bank.

CreditScore can have an effect on customer churn, since a customer with a higher credit score is less likely to leave the bank.

Age this is certainly relevant, since older customers are less likely to leave their bank than younger ones.

Tenure refers to the number of years that the customer has been a client of the bank. Normally, older clients are more loyal and less likely to leave a bank.

Balance also a very good indicator of customer churn, as people with a higher balance in their accounts are less likely to leave the bank compared to those with lower balances.

Churn prevention allows companies to develop loyalty programs and retention campaigns to keep as many customers as possible.

Machine learning process

- Load the data into Python.
- Clean data. Deal with missing values.
- Visualize the data, by creating some graphs.
- Split data set into 80% training set and 20% test set.
- Train the machine learning algorithm. Typically multiple algorithms are tried.
- Evaluate the model, using accuracy, false positives, false negatives.

Until recently standard machine learning algorithms gave the best performance, but foundational tabular models (based on neural networks) are starting to perform the best for small data sets.

Real world applications

Data Science projects at a University.

- The input data will be an Excel file from a public repository such as Kaggle.
- A typical data science project will produce a "notebook" of code that will implement the classifier.

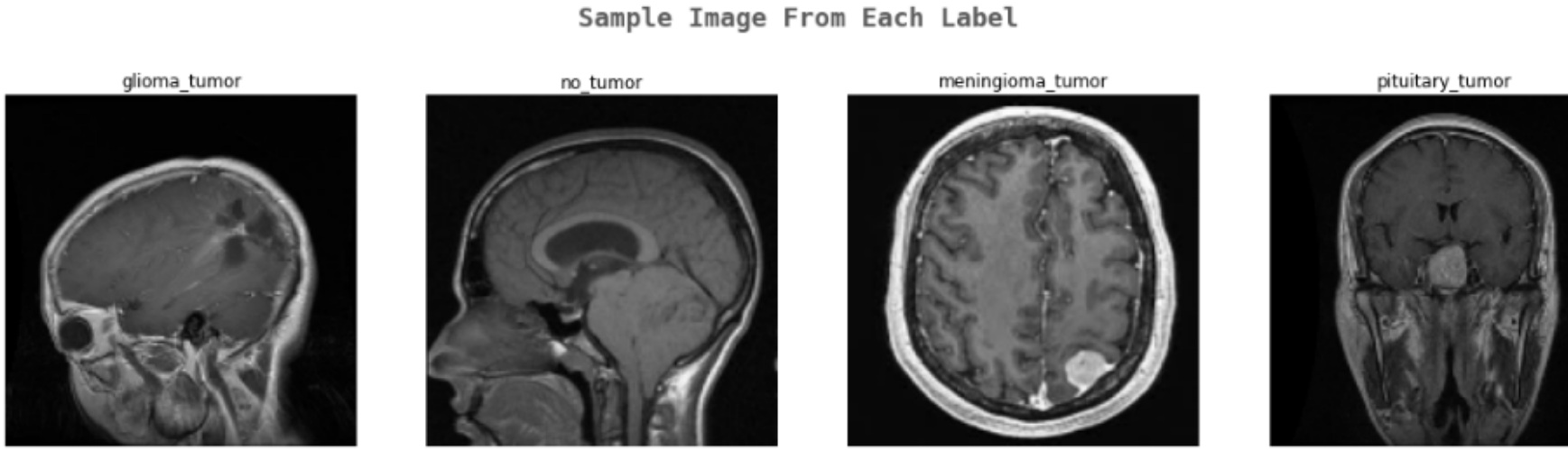
Data Science projects in industry

- The code needs to be supplied to people (perhaps working in a call center) who don't know programming, so the code has to be wrapped in an App.
- The input data will come from a database of some kind.
- There could be model drift, because of for example, an advertising campaign of the company or rivals.
- It should be able to quantify the savings from using the model (ROI Return On Investment).

It is very hard to get "real data" from companies.

Image classifier

- A dataset of MRI images by experts labeled by experts.
- Build a machine learning algorithm to reproduce the labels of the images.



- On the computer the images are just a table of numbers.
- The images are a 2D array of numbers that are inputted to the neural network.

Will AI replace radiologists

In 2016, Professor Geoffrey Hinton stated that

> *People should stop training radiologists now and "it is just completely obvious that within 5 years deep learning will do better than radiologists.*

Radiologists are still needed!

> *At Johns Hopkins Medicine, all mammogram screenings are reviewed by fellowship-trained, specialized breast radiologists, with the support of AI software at no additional charge. The purpose of the AI review is to improve the accuracy of mammogram results.*

- AI is also particularly good at picking up small and subtle distortions, which could be early indicators of suspicious findings, prompting clinicians to recommend a needed biopsy.

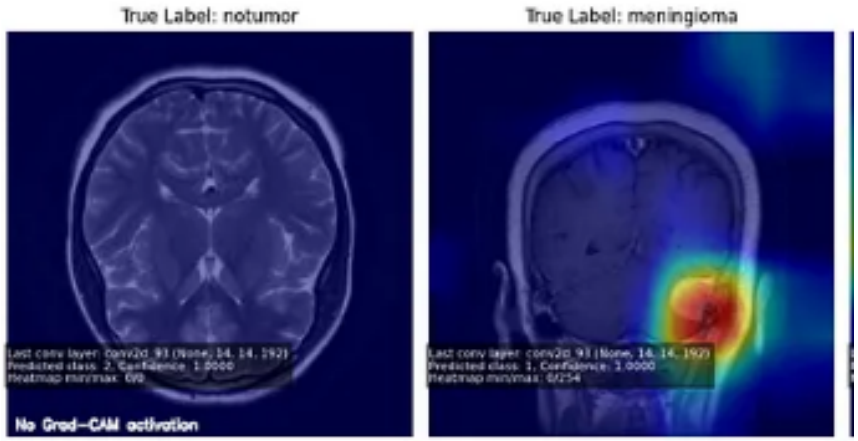
Explainability

Consultant, I am sorry to say the algorithm has diagnosed that you have a disease.
Patient, Why?
Consultant, the algorithm said so.

- There are techniques, known as explainability, to try to understand the output from machine learning algorithms.

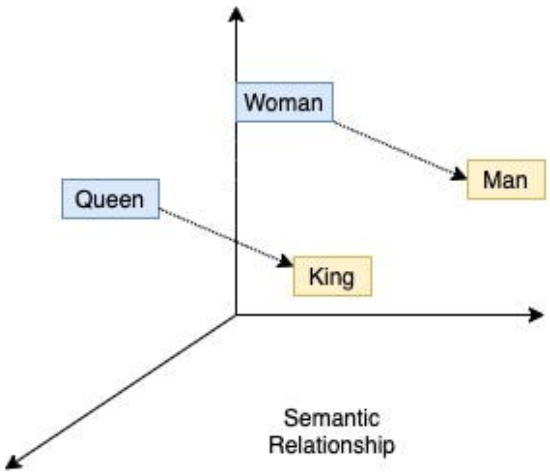
Explainability of classifier Brain Tumor MRI

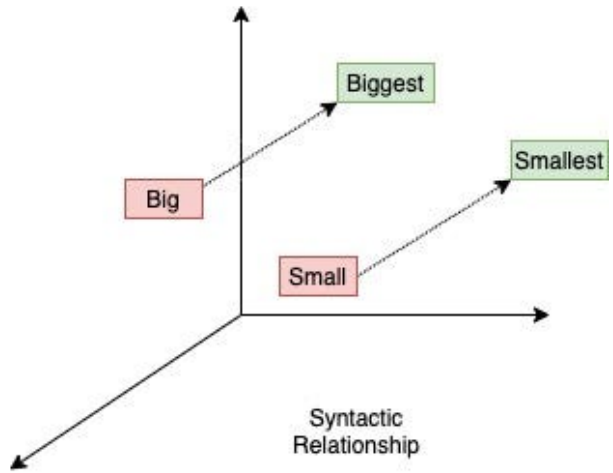
- One algorithm for explainability is Grad-CAM.
- Areas highlighted in red indicate regions the model focuses on.
- This is subtle because a many layered network is used in the classifier.



The use of text with deep learning

- To use text with Neural Networks, the words need to be converted to vectors (collection of numbers [0.1, 0.2, -0.3]).
- The word embeddings capture some meanings.





Introduction to Artificial General Intelligence

Artificial General Intelligence (AGI) is not a well defined term.
In fact, even intelligence is not a well defined term.

IQ Set of standardizes tests. Normalized so the mean IQ for the population is 100%.

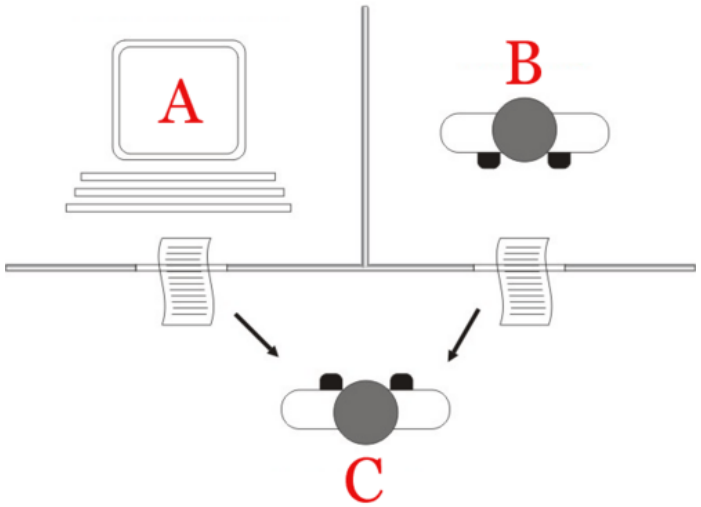
- g-factor, common intelligence factor.
- The book, "The Mismeasure of Man", by Stephen Jay Gould, criticizes the ideas of IQ.

Multiple intelligences Theory developed by Howard Gardner.

- eight intelligences, including linguistic, logical-mathematical, spatial, musical, bodily-kinesthetic, interpersonal, intrapersonal, and naturalistic.
- Evidence is not clear for multiple intelligences.

Turing test

Introduced by Alan Turing as the "imitation game" in his 1950 paper "Computing Machinery and Intelligence", the test involves a human evaluator asking questions to another human and a machine via a computer keyboard and monitor. If the evaluator cannot tell from the written responses which is the human and which is the machine, then the machine has passed the Turing test.



Artificial General Intelligence

- It is an interesting to research into building computer models of intelligence and thought.
- However, the US big technology companies, such as OpenAI, are not developing AGI to understand the human mind. They mostly want cash (or to be dominated/destroyed by AGI).



Artificial General Intelligence

Dario Amodei, wrote an essay, "Machines of Loving Grace" in 2024.

`https://darioamodei.com/essay/machines-of-loving-grace`

- In terms of pure intelligence, AGI is smarter than a Nobel Prize winner across most relevant fields – biology, programming, math, engineering, writing, etc. This means it can **prove unsolved mathematical theorems**, write extremely good novels, write difficult codebases from scratch, etc.
- AGI does not just passively answer questions; instead, it can be given tasks that take hours, days, or weeks to complete, and then goes off and does those tasks autonomously, in the way a smart employee would, asking for clarification as necessary.

We could summarize this as a "country of geniuses in a datacenter.

Quiz



AGI and health

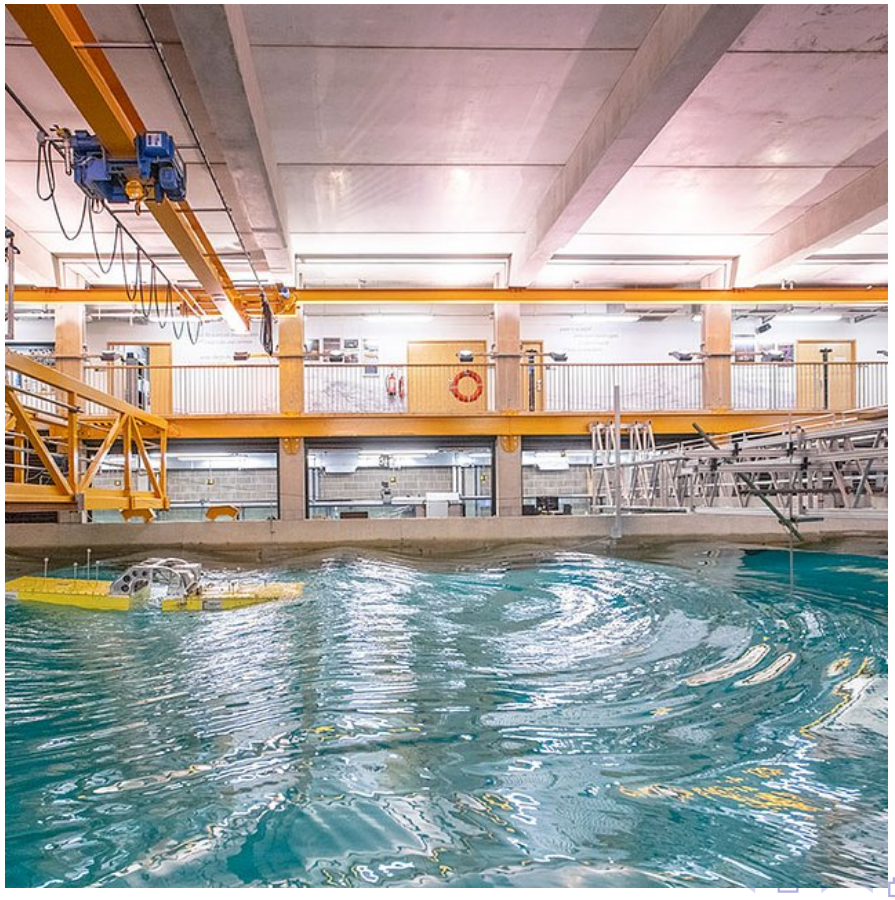
From Dario Amodei's essay "Machines of Loving Grace". Some advantages of AGI in healthcare:

- Elimination of most cancer.
- Very effective prevention and effective cures for genetic disease.
- Prevention of Alzheimer's.
- Doubling of the human lifespan.

And people complain about data centers.

Will AGI solve the climate crisis?

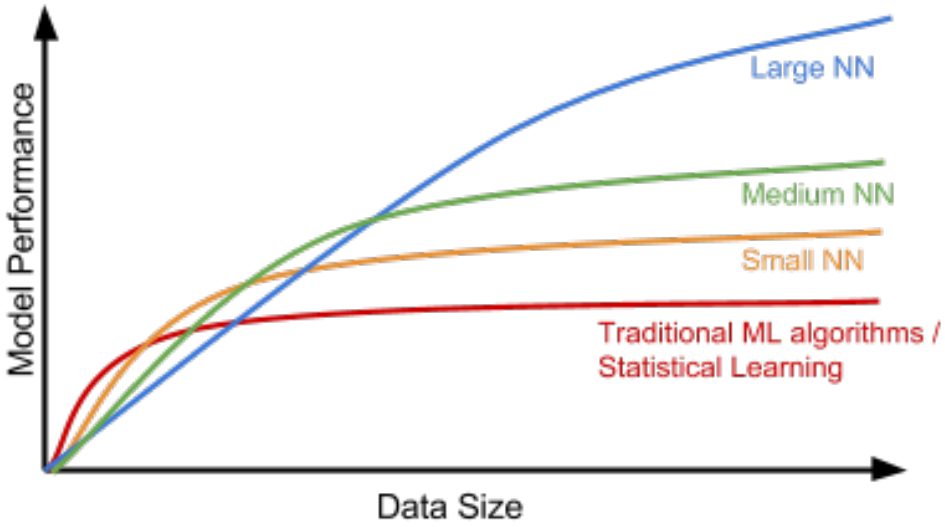
- The Ocean Basin, at the University of Plymouth, is a facility that allows waves and currents to be generated at any relative orientation and can be run at different water depths.
- Computer simulation on its own is not enough to design floating turbines.



Why are so many data centers being built?

Below is a cartoon of performance of AI with the size of data.

- They are building more data centers and getting more data because the scaling analysis suggested that the performance would improve with more compute.
- It may be that the performance of LLMs have now plateaued. New ideas from symbolic AI or world models may be needed (Gary Marcus).



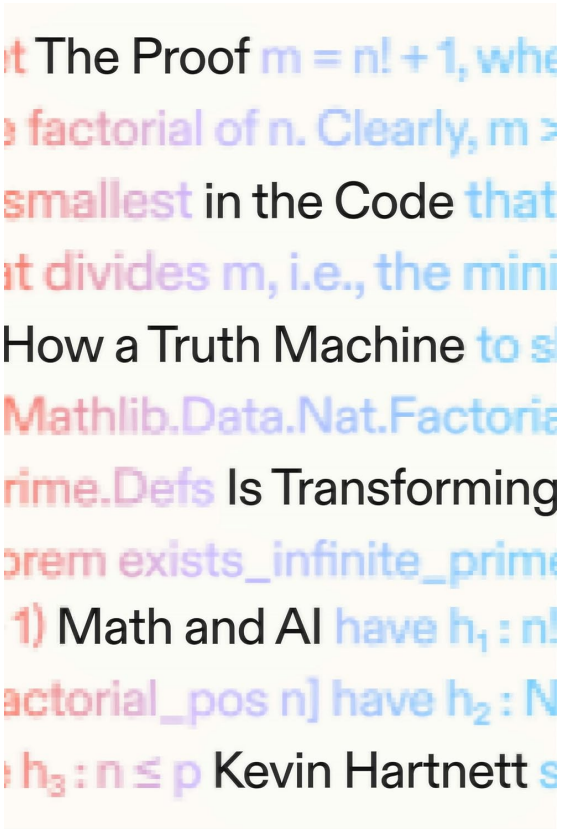
AI and the crisis in pure mathematics

- Pure Mathematics describes the study of mathematical concepts independently of any application outside mathematics.
- (Pure mathematics does has applications, such as Cryptography.)
- A key part of pure mathematics is the idea of proof. (For example, there are an infinite number of prime numbers.)
- As mathematics proofs got more complicated, it is getting harder to check the proofs.
- Mathematicians and computer scientists have developed computer systems to check proofs. Essentially a proof is written as a computer program.

The most popular language to check proofs is **lean**.

History of lean

- The book below gives a good history of the development of lean.
- There are libraries of pure mathematics.



Example of lean code

generate a simple lean program

Here is a simple Lean 4 program that defines a custom addition function, evaluates it, and proves a basic theorem about natural numbers.

Lean

```
-- Define a simple function that adds two natural numbers
def add (a b : Nat) : Nat :=
  a + b

-- Evaluate the function
#eval add 3 5 -- Output: 8

-- Prove a simple theorem: adding zero to a number leaves it unchanged
theorem add_zero_eq (n : Nat) : n + 0 = n := by
  rfl
```

Book: Maths Proofs in Lean: First Steps by Tariq Rashid.

AI and pure mathematics

- The conversion of proofs in pure mathematics into a lean program is an ideal case for AI to work with.
- In particular the lean proof can be checked, if it compiles.
- This stops the problem with LLM hallucinations.

Solving pure maths problems is hard, but curing cancer, or stopping climate change will be much harder.

The Millennium Prize Problems

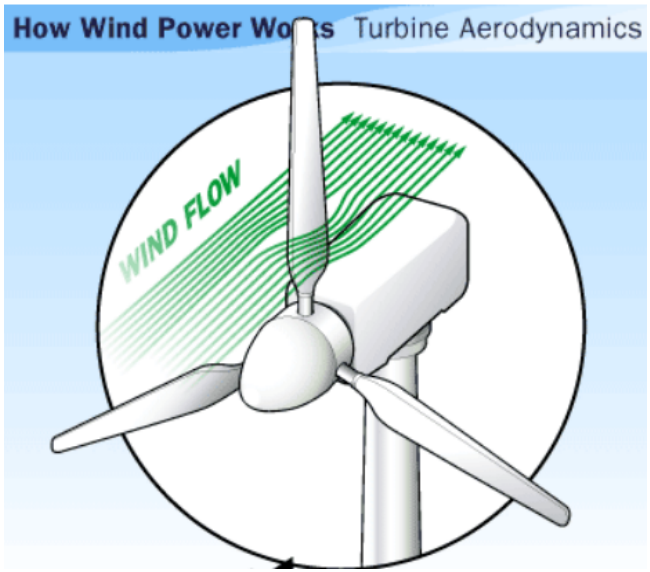
> *The Clay Mathematics Institute is a global organization dedicated to furthering the beauty, power and universality of mathematical thought.*

The Clay Mathematics Institute of Cambridge, Massachusetts (CMI) established seven Prize Problems in Mathematics.

- Prizes were established in 2000.
- Each prize is worth 1,000,000 $.
- One problem is about the Navier-Stokes Equation.

Navier-Stokes Equation

- Navier-Stokes Equation is the equation which governs the flow of fluids such as water and air.
- The Navier-Stokes Equation is regularly solved numerically using supercomputers.
- No proof has been found answering the most basic questions one can ask: do solutions exist, and are they unique.



Navier–Stokes priority controversy

- (**Breakthrough**) On 8 September 2026, OpenAI proposed a solution to the Navier–Stokes existence and smoothness problem with a smooth external force.
- This required an **88-hour run** , an multiagent system of OpenAI produced a 166-page paper along with a computer-verified Lean formalisation.
- Cost of the proof around 5,000,000 $.
- It will probably not be submitted to the Clay institute to claim the prize.

Two mathematicians were already working on the problem using LLMs. They felt their work had helped the OpenAI agents, because their work had used LLMs and the prompts were sent back to OpenAI.

Other mathematical results

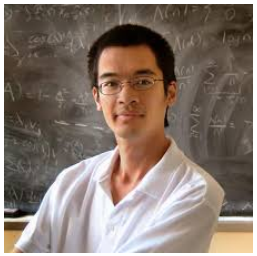
- On 6 October 2026, OpenAI released 722 manuscripts claiming to have solved a large number of open and unsolved major mathematical problems
- The release was done by uploading the results to a GitHub repository. This is a non-standard way to publish new results.

Summary AI and pure mathematics

- Mathematicians are not happy with the AI generated proofs.
- The AI systems are producing proofs, but no insight.

  As machines race ahead in solving problems, mathematicians must establish a new era that prizes human understanding over simply publishing more solutions.

Terence Tao

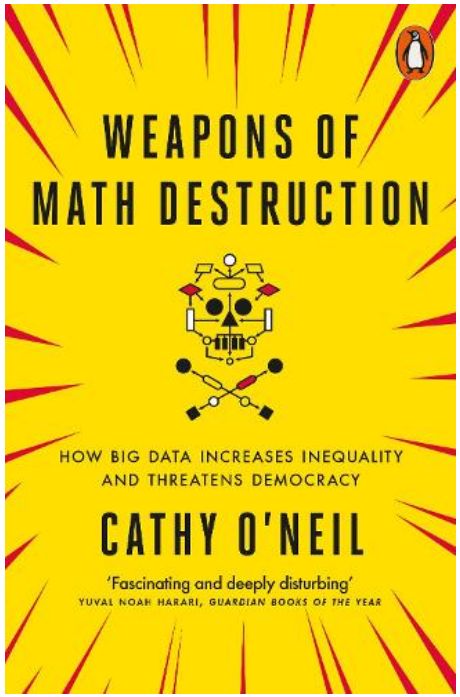


Note, that mathematical proofs are a much smaller part of computer simulations and applied mathematics.

AI alignment problem

- The AGI may have goals than the human race. (AI alignment problem)
- There are many ethical concerns with the use of AI, before we have to worry about a rogue AGI.

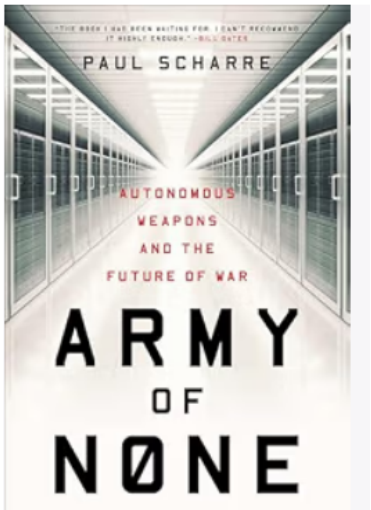
Book published in 2017



Will AGI kill us?

Before we should worry about a rogue AI wiping out all humanity.

- In 1988, Iran Air Flight 655 was shot down by two surface-to-air missiles fired by USS Vincennes, a United States Navy warship. This caused 290 deaths.
- Partly the "mistake" was technological issues.



Conclusions

- AI was already being used in mathematics and physics before ChatGPT burst upon the world in 2022.
- AI is a useful tool. For example: churn prediction and medical image classifiers.
- I believe that Silicon Valley's search for AGI is a distraction.
- One issue is the cost of AI use.
- A critical issue is access to datasets. The UK should not give the US tech companies (especially Palantir) access to NHS data.

It is not clear that AI will produce a vastly different economy.